Akoizo Original Practice - 2027 Rotation Form 2

Division C | Anatomy And Physiology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Type II pneumocytes produce:

Unit: Lungs, Alveoli, Pleura, and Breathing Muscles.
Worked-case data: At one instant, alveolar pressure is 0 cm H_2O relative to atmosphere and pleural pressure is -7 cm H_2O. At another, alveolar pressure is -1 and pleural pressure is -6.

A. HCl

B. Antibodies

C. Mucus

D. Surfactant

Answer: _____

2. (2 points) Exercising muscle unloads more O2 because of:

Unit: Gas Exchange, Hemoglobin, and pH Control.
Worked-case data: Compare blood with hemoglobin 15 g/dL versus 8 g/dL. Both samples have saturation 0.98 and PO_2 100 mmHg.

A. High pH

B. Left shift from alkalosis

C. Right shift from acid, CO2, heat

D. Low 2,3-BPG

Answer: _____

3. (2 points) Apical cavitary disease with caseating granulomas suggests:

Unit: Infection and Restriction: Pneumonia, TB, Fibrosis.
Worked-case data: Case A has a focal air-space opacity and ongoing blood flow through poorly ventilated tissue. Case B has diffuse interstitial thickening and reduced total lung capacity.

A. Asthma

B. TB

C. Emphysema

D. Pneumothorax

Answer: _____

4. (2 points) B12 is absorbed in the:

Unit: Small Intestine, Liver, Pancreas, and Absorption.
Worked-case data: A case supplies normal pancreatic enzyme production but greatly reduced bile delivery to the small intestine.
Predict the most directly affected step.

A. Jejunum

B. Terminal ileum

C. Colon

D. Duodenum

Answer: _____

5. (2 points) Lactose intolerance is caused by deficiency of:

Unit: Lactose Intolerance, Obesity, and Exercise Metabolism.
Worked-case data: A teaching model supplies 24 g lactose during transit. In one setting, 18 g can be hydrolyzed; in another, only 6 g can be hydrolyzed.

A. Maltase

B. Pepsin

C. Sucrase

D. Lactase

Answer: _____

6. (2 points) Hard fixed nontender supraclavicular node most suggests:

Unit: Spleen, Nodes, and Barriers in Defense.
Worked-case data: One case involves particles arriving from a skin wound in lymph. Another involves microbes circulating in blood. A third involves intestinal luminal exposure.
Separate investigation observations:
Input: The organ receives and filters blood.
Architecture: White pulp supports immune responses; red pulp processes erythrocytes.
Function: The case emphasizes defense against blood-borne material.

A. Vaccine reaction

B. Viral URI

C. Malignancy

D. Allergy

Answer: _____

7. (2 points) CD8 T cells recognize antigen on:

Unit: Adaptive Immunity, Antibodies, and Allergy.
Worked-case data: An experiment shows a delayed small antibody rise after first exposure and an earlier larger rise after a later exposure to the same antigen.

A. MHC II only

B. MHC I

C. Free antibody

D. Complement

Answer: _____

8. (2 points) Heat-worsened blurred vision with INO suggests:

Unit: Autoimmunity, MS, RA, and Systemic Allergy.
Worked-case data: Case A describes central myelin injury. Case B describes destructive synovial inflammation. Case C describes rapid systemic effects after re-exposure to an external antigen.
Separate investigation observations:
Timing: Effects develop rapidly after re-exposure.
Trigger: The trigger is an external antigen encountered previously.
Mechanism: Mast-cell mediators alter vascular permeability and airway smooth muscle.

A. MS

B. Gout

C. Celiac

D. RA

Answer: _____

9. (2 points) Normal pleural pressure during quiet breathing is:

Unit: Lungs, Alveoli, Pleura, and Breathing Muscles.
Worked-case data: At one instant, alveolar pressure is 0 cm H_2O relative to atmosphere and pleural pressure is -7 cm H_2O. At another, alveolar pressure is -1 and pleural pressure is -6.

A. Negative around -4 mmHg

B. Equal to alveolar

C. Positive 10 mmHg

D. Atmospheric

Answer: _____

10. (2 points) Most CO2 is carried as:

Unit: Gas Exchange, Hemoglobin, and pH Control.
Worked-case data: Compare blood with hemoglobin 15 g/dL versus 8 g/dL. Both samples have saturation 0.98 and PO_2 100 mmHg.

A. Myoglobin

B. Dissolved gas

C. Carbamino only

D. Bicarbonate

Answer: _____

11. (2 points) A teaching spirometry case gives tidal volume 540 mL, dead space 150 mL and respiratory rate 12/min. Find alveolar ventilation.

Answer: _____
_____
_____

12. (2 points) Use the same case to calculate minute ventilation: tidal volume 540 mL and rate 12/min.

Answer: _____
_____
_____

13. (2 points) A model narrows an airway while pressure difference and other conditions are fixed. Explain the likely airflow consequence.

Answer: _____
_____
_____

14. (2 points) Describe how pleural fluid assists ordinary breathing.

Answer: _____
_____
_____

15. (2 points) Why does the small intestine combine villi, microvilli and a thin absorptive epithelium?

Answer: _____
_____
_____

16. (2 points) Distinguish the digestive roles of bile and pancreatic lipase.

Answer: _____
_____
_____

17. (2 points) Explain why a pancreatic bicarbonate secretion can assist digestion of acidic stomach contents.

Answer: _____
_____
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18. (2 points) Compare innate and adaptive immune responses without calling either one universally faster or stronger in every situation.

Answer: _____
_____
_____

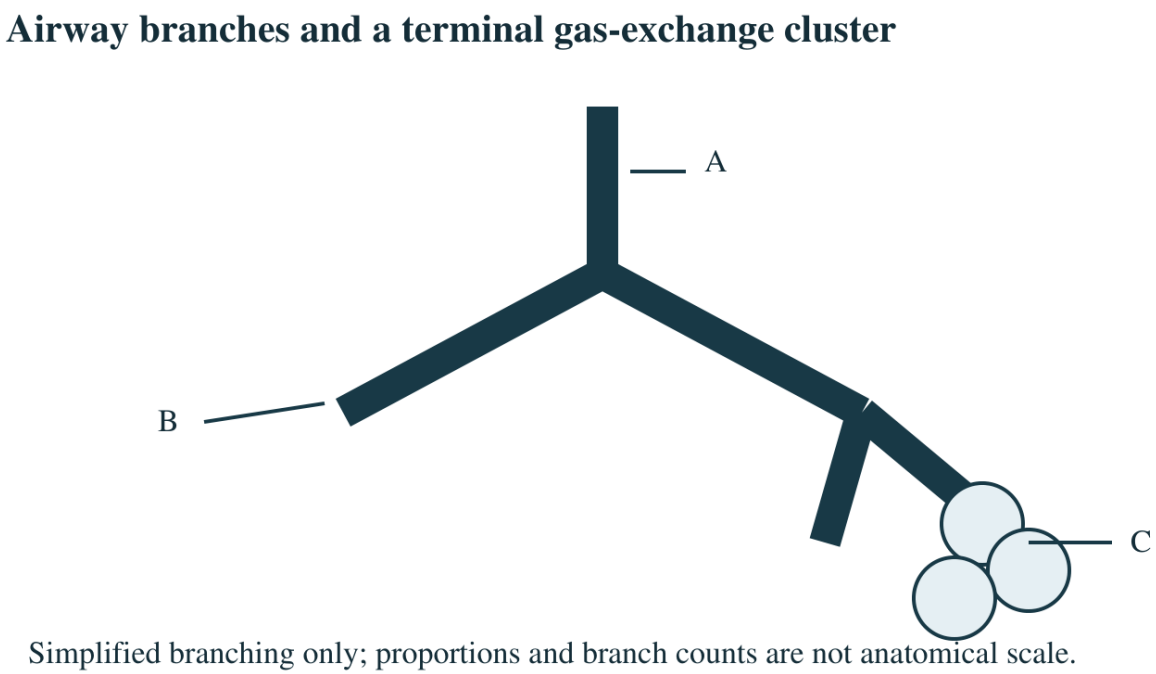
19. (2 points) Explain why a secondary antibody response can differ from a primary response.

Answer: _____
_____
_____

20. (2 points) In a teaching case, helper T-cell activation is impaired. Give one potential consequence for a coordinated adaptive response.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Name the structure labeled A.



F1. Respiratory branching schematic (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Which labeled region most directly provides the thin gas-exchange interface?

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

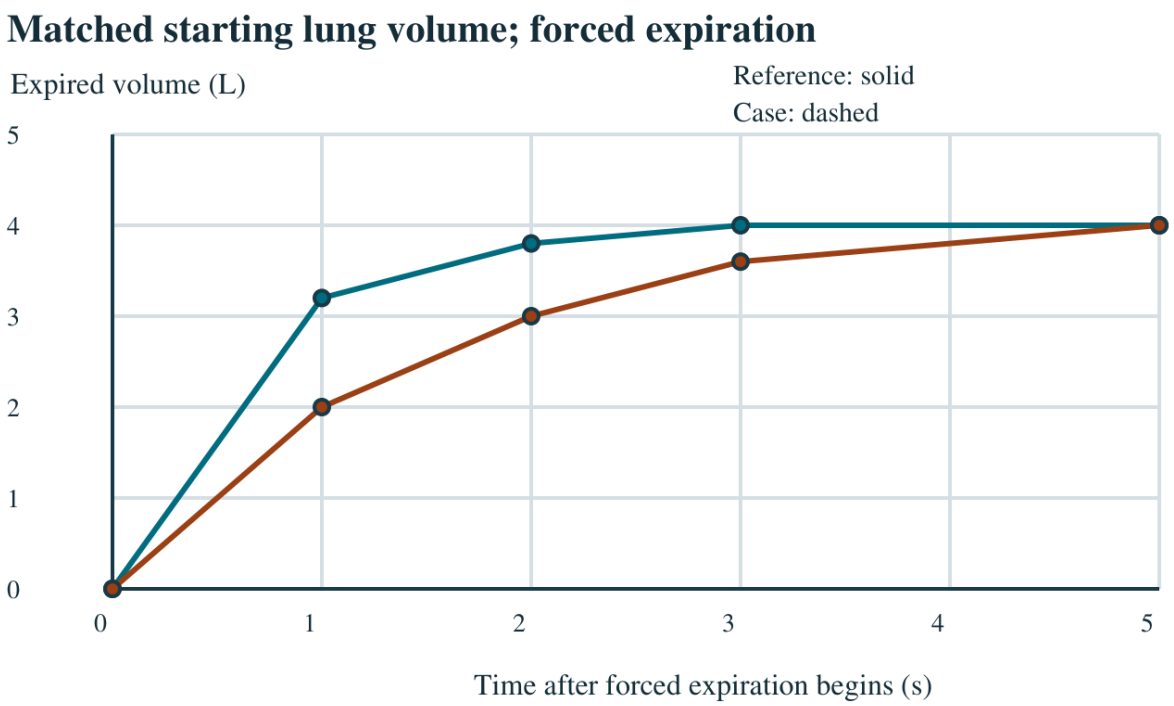
Answer: _____
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23. (2 points) Use Figure F1. Trace inspired air from A toward C using the branching arrangement.

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Calculate case FEV1/FVC as a percentage.



F2. Forced expiratory volume curves (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Which curve shows slower early emptying, and what pattern does that support studying?

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

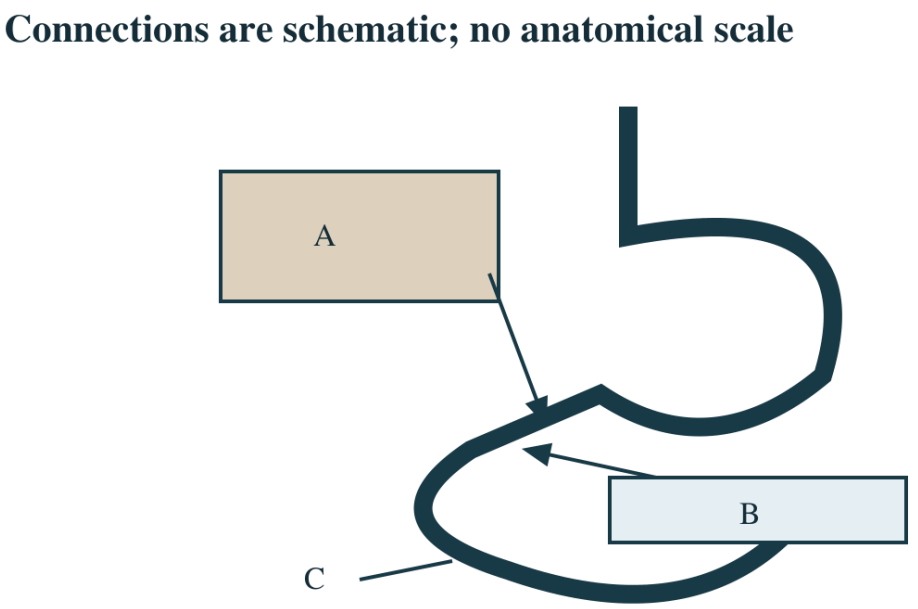
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Find reference-minus-case expired volume at1 s.

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Which labeled organ produces bile? Give its name.



A large accessory organ; B lobulated gland; C coiled intestinal region

F3. Digestive organ relationships (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Name B and explain one digestive secretion it delivers toward the shown intestinal connection.

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

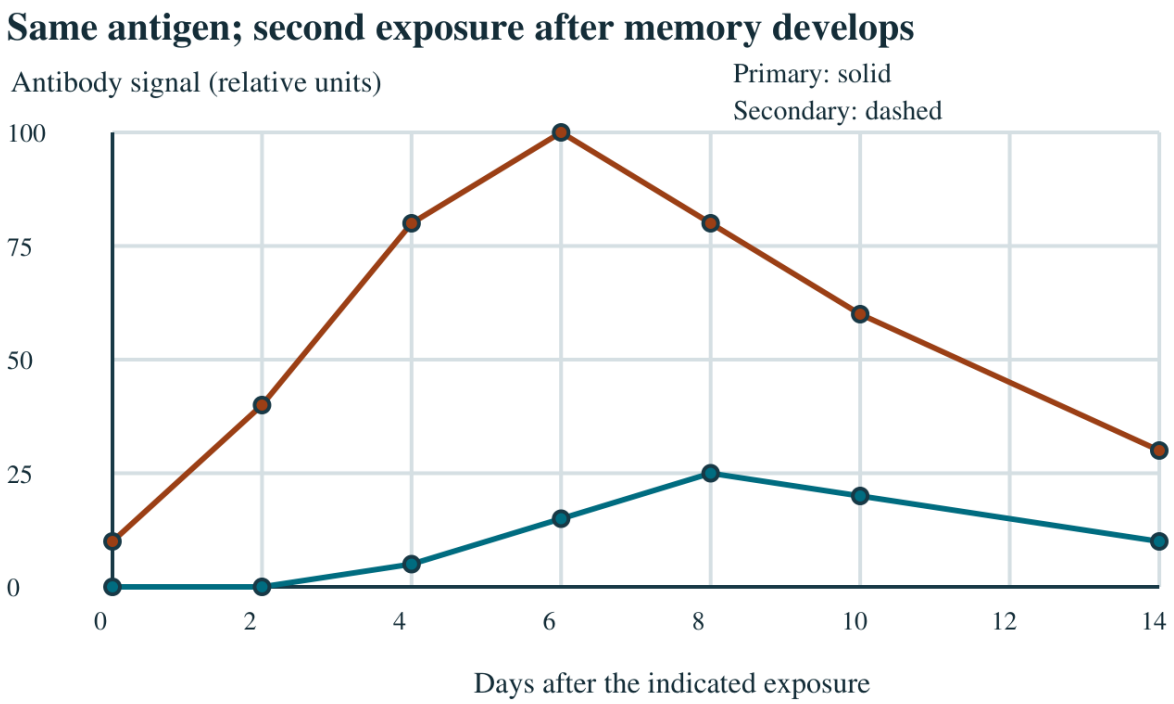
Answer: _____
_____
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29. (2 points) Use Figure F3. Which labeled region is most directly associated with most nutrient absorption?

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Read the ratio secondary-peak / primary-peak signal.



F4. Primary and secondary antibody responses (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. At day4, calculate secondary-minus-primary signal.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Explain how the curve comparison is consistent with adaptive immune memory.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____